



Instructor: Date:

Learner: Regional Branch:

Program: Center:

CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Different Ways to Compose a System



Learning Outcome:

Build the application installation program, using IT development tools available in the market, based on solution architecture features. Recognizes and implements the different types of architecture styles that can be included in the IT solution.

Can Do Statements:

- I can talk about architecture styles.
- I can describe the pipes and filters architecture style.
- I can explain the object-oriented style.

Ask your partner:

- What types of architecture styles do you know about?
- What aspects would you take into account to design your own application?





Let's Explore!

Drake needs to know about architecture styles.



Instructions: Read the text below.

Sara: Hey Drake, what are you reading?

Drake: It's a book I got to learn about architecture styles, I have to design an application as part of a project in my engineering class.

Sara: Cool! What have you learned about architecture styles so far?

Drake: Architecture styles are patterns or frameworks for software development. They are designed to solve difficulties when structuring a program. These styles are implemented to define the relation among the components that will be a part of the system. They are very important because we may establish objectives and restrictions, so, we don't fall in a deadlock.

Sara: And what style are you going to use?

Drake: There's one called pipes and filters, it's very simple. A filter is structured to read a stream of data as input and produce output, filters might have parameters to perform this action. A pipe is the connection that carries the data to the filters; it serves as a conduit of outputs that become inputs once they're delivered to the filter.

Sara: You might know a lot about software to implement something like that.

Drake: Yes, I'm very interested in these concepts. Actually, there's another architecture style that I like. It is called object-oriented and it's about breaking down a system into individual parts. They may use reusable components that communicate with one another and build a constant throughput by sending and receiving messages.

Sara: Well, you might take into account that not all systems can be broken down into defined objects. I worked on a similar project where I used the object-oriented style, and you may need to verify features such as inheritance, encapsulation, abstraction, and polymorphism.

Drake: Wow, I didn't expect you to know about this topic. Would you help me understand some concepts?

Sara: Sure!



**Let's Practice!**

This topic is easy to understand!

Instructions: Read the sentences. Write the missing word in the blank. Use: are required, is structured, are carried, is called, are used.

1. There are different styles, and one of them ____ pipes and filters.
2. A filter ____ to read a stream of data as input and produce output.
3. Some parameters ____ in filters to perform actions to read and transmit the information.
4. Data ____ by the pipes that connect everything with the filters.
5. Pipes ____ as a conduit of outputs that become inputs once they're delivered to the filter.

**Let's Practice!**

Sara has used a different kind of style.

Instructions: Underline the correct answer. Read the sentences aloud. Check your pronunciation with your instructor.

1. There's another architecture style ____ object-oriented.
a. that is called
b. that is calling
c. which is call
d. that are calling
2. An object-oriented ____ breaking down a system into individual parts.
a. are relating to
b. are related to
c. is related to
d. are relate to
3. All individual parts ____ reusable components that communicate with one another.
a. is uses as
b. is use such
c. is used as
d. are used as
3. A constant throughput ____ sending and receiving messages through those components.
a. are done by
b. is built by
c. is gone by
d. are read by
4. Features like inheritance, encapsulation, abstraction, and polymorphism ____ this kind of style.
a. is verified with
b. are verify with
c. are verified with
d. None of the above





Extension Activity

You are going to pick out the best option.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about architecture styles
- Takes notes about those architecture styles
- Make a chart where you write a little description of each architecture style.
- Use the information in the chart to create a brochure to promote the style that somebody could use.
- Share your brochure with your partners.

Self-Assessment

Yes

No

- I search information about architecture style. ☐ ☐
- I describe each architecture style. ☐ ☐
- I express what style could be the best option to use in a system. ☐ ☐
- I share with my partners. ☐ ☐



Target Words

- **conduit** (noun): where components travel through to get from one filter to another
e.g. It serves as a conduit of outputs that become inputs once they're delivered to the filter.
- **throughput** (noun): quantity of processes within a specific time
e.g. They may use reusable components that communicate with one another and build a constant throughput.
- **stream** (noun): a flow of components
e.g. A filter is structured to read a stream of a data as input and produce output.
- **pipe** (noun): means of transport for data
e.g. A pipe is the connection that carries the data to the filters.
- **filter** (noun): component with a set of inputs and outputs
e.g. Filters might have parameters to perform this action.
- **deadlock** (noun): a stage where a process is not able to continue
e.g. We may establish objectives and restrictions, so, we don't fall in a deadlock.



Learning Points

Modal verbs for speculating- **might**/ **may**.

You use the modal verbs **might** and **may** to create statements for speculating about any topic.

The modal verbs **may** and **might** besides expressing probability, they are used to speculate about procedures or reactions.

Let's check the grammatical structure:

Subject + **may/might** + verb (base form) + complement.

Example:

*Software architectures **may** include useful data structures.*

*Architectural descriptions **might** be part of the style definition.*

Although **may** and **might** are very similar in meaning, their difference lays on the intention of the message. If you use **may** in a statement, it is because you are surer about it than if you use **might**, because, in the latter case, you are less sure about the situation.

Examples

- *Filters **might** have parameters to perform this action.*
- *We **may** establish objectives and restrictions, so, we don't fall in a deadlock.*
- *You **may** need to verify features such as inheritance, encapsulation, abstraction, and polymorphism.*



Learning Points

Architecture styles Summary.

The structure of any system formed by software components is called **architecture**.

An architectural style determines the vocabulary that is going to be used for the components and the connectors within a system. There are different architectural styles that define how a particular system is designed. So far, there is not a global taxonomy to develop software architectures, instead there are several individual ways to create them.

Now, we are going to check some of the most common styles where architecture is represented.

Examples

- *Architecture styles are patterns or frameworks for software development. They are designed to solve difficulties when structuring a program.*
- *These styles are implemented to define the relation among the components that will be part of the system.*
- *There's another architecture style that I like. It is called object-oriented and it's about breaking down a system into individual parts.*